

SPACE TRAFFIC MANAGEMENT · HACKATHON SUBMISSION

# Orbital Sentinel

From two-line elements to a cascade-safe avoidance burn — every number computed, cross-checked and traceable.

**501**

objects tracked live (SGP4)

**170**

conjunction alerts in 24 h (1 critical)

**18/18**

physics cross-checks passing

**0.962**

ML surrogate F1 @ Pc 1e-4 (held-out)

**3/501**

objects with unknown agency

**149/149**

independent maths checks passing

Live backend snapshot: simulation epoch 2026-10-08 09:56 UTC (fetched 2026-10-08 09:57 UTC). Demo scenario run: 2026-10-08 09:38 UTC. Report generated 2026-10-08 09:58 UTC by docs/report/build\_report.py.

Repository: orbit-sentinal · FastAPI backend + React/Cesium frontend · deep-dive: docs/HOW\_IT\_WORKS.md

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## 00

## Executive summary

the problem, what we built, and the five results that matter

**The problem.** Low Earth orbit holds tens of thousands of catalogued objects. Operators need to know *which* close approaches are dangerous, *what* to do about them, and whether the fix creates a new problem. A single collision produces hundreds of fragments that threaten other satellites, sometimes owned by other agencies (the Kessler cascade).

**What we built.** Orbital Sentinel runs the whole chain on a simulation clock: SGP4 propagation → 24-hour conjunction screening → Foster collision probability with a TLE-age covariance → an explainable ML surrogate (advisory) → a single, documented decision score → re-propagated, cascade-safe avoidance burns costed on 12 real engines → a NASA Standard Breakup Model debris cloud → fragment screening → a cascade graph mapped to owning agencies through the CelesTrak SATCAT. A Cesium globe shows it live.

### Five headline results

- ▶ **Screening is a real look-ahead.** 501 objects are screened over 24 h at 60 s with a KD-tree and Brent-refined TCA; this snapshot yields 170 pairs (1 critical, 7 warning). Screening 500 objects takes 0.71 s in-process on a laptop; a full API pipeline recompute took 41.7 s during this report run.
- ▶ **Collision probability is cross-validated.** 18 of 18 live physics checks pass (Foster vs Chan series, Monte Carlo, exact Rice CDF and Alfano maximum; propagation energy; CW, Hohmann and rocket-equation manoeuvre checks; NASA SBM fragment counts).
- ▶ **The ML surrogate is measured, explained and kept in its place.** Held-out MAE 0.106 dex vs 1.57 dex for a miss-distance-only baseline; precision 0.975 / recall 0.950 at  $P_c 1e-4$ . TreeSHAP shows the main driver is *Older TLE age*. PCA was tested and rejected with numbers. The model never overwrites the physics  $P_c$ .
- ▶ **Avoidance is cascade-safe and costed.** For the demo encounter ( $P_c 4.75 \times 10^{-5}$ ) the planner evaluated 48 burns, re-screened shortlisted options against the whole catalogue, and chose **+S 0.5 m/s** (new  $P_c 7.70 \times 10^{-11}$ , 0.070 kg propellant, engine Aerojet AMBR); the CW analytic drift agrees with propagation to 0.32 %.
- ▶ **Collision → debris → cascade is end-to-end.** Breaking up the demo pair at its predicted TCA gives 1104 fragments  $\geq 10$  cm (EMR 34280 J/g, catastrophic); screening them produced 8 fragment alerts on 8 satellites, wired into the cascade graph and the owning agencies.

#### Design principle

physics decides → ML explains and cross-checks → the decision score combines both, transparently

Every assumption (mass, covariance growth, hard-body radius,  $I_{sp}$ ) is a named constant and each output carries a provenance tag (mass\_source, sigma\_source, hbr\_source, isp\_source). Section 11 lists what the system does *not* do.

## 01

## System pipeline

what runs on every 30 s refresh and on a simulated collision

## WHAT IT IS

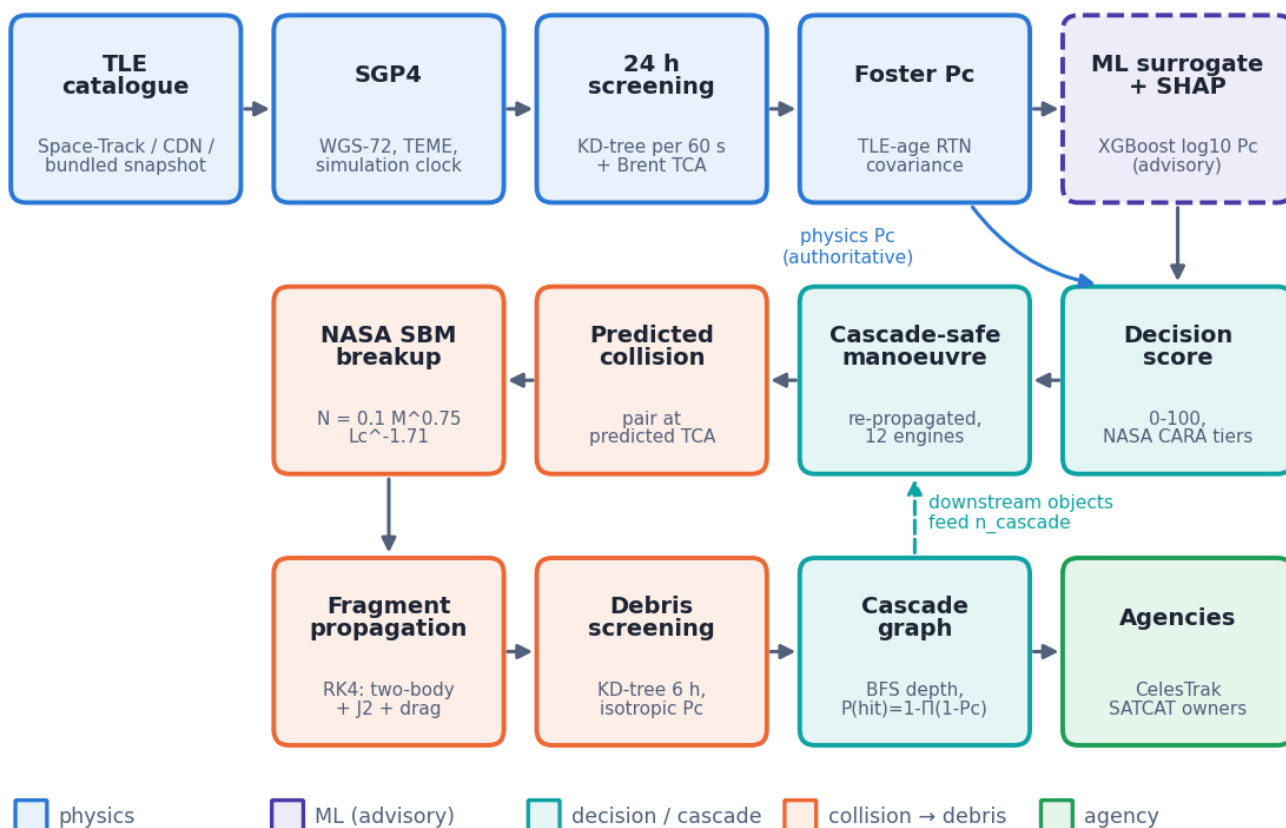
One backend loop (refresh\_alerts\_once) that turns TLEs into alerts, decisions and burns; a second path turns a predicted collision into a debris cloud and new alerts.

## HOW IT'S COMPUTED

Each box is a module with its own tests: SGP4 (sgp4 lib), screening.py, pc\_methods.py, ml/risk\_api.py, decision.py, maneuver\_planner.py, propulsion.py, breakup.py, debris\_model.py, cascade\_planner.py, satcat.py.

## WHY TRUST IT

All physics runs on one simulation clock; the ML branch is advisory and cannot lower the physics action; outputs carry provenance tags so judges can see which numbers are assumptions.



Source: Pipeline as implemented in backend/main.py::refresh\_alerts\_once and the /api/debris/simulate route (see docs/HOW\_IT\_WORKS.md for each module). Dashed box: advisory model that never writes the physics Pc.

## Stage-by-stage key equations

| Stage     | Core relation (as implemented)                                                                                                                 |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------|
| SGP4      | TEME position/velocity from TLE; executed burns: $r(t) = r_{SGP4}(t) + [r_{j2}(t; x_0 + \Delta v) - r_{j2}(t; x_0)]$                           |
| Screening | KD-tree radius = threshold + $v_{rel,max} \cdot \text{step}/2$ ; $t^* = -(\Delta r \cdot \Delta v) /  \Delta v ^2$ ; Brent refine (xatol 1 ms) |
| Foster Pc | $P_c = \iint_{ x  \leq HBR} N(x; b, C_B) dx$ , $C_B = B(C_1 + C_2)B^T$                                                                         |
| Surrogate | XGBoost $f(7 \text{ features}) \approx \log_{10} P_c$ ; base + $\sum \text{SHAP}_i$ = prediction                                               |
| Decision  | score = $100 \sum w_i n_i$ ; action from physics Pc tiers $1e-4 / 1e-5 / 1e-7$                                                                 |
| Manoeuvre | $m_p = m_0(1 - e^{-\Delta v / (I_{sp} g_0)})$ ; $t_{burn} = m_p I_{sp} g_0 / F$                                                                |
| Breakup   | $N(\geq L_c) = 0.1 M^{0.75} L_c^{-1.71}$ ; catastrophic if $EMR \geq 40 \text{ J/g}$                                                           |
| Cascade   | $P(\text{hit}) = 1 - \prod(1 - P_{c,i})$ ; depth = BFS hops from the collision event                                                           |

## 02 Catalogue & agencies

who is up there, where, and who owns it

### WHAT IT IS

501 objects propagated live, attributed to an owner; a 30,482-row SATCAT snapshot gives owner, object type, RCS size and orbit for the whole catalogue.

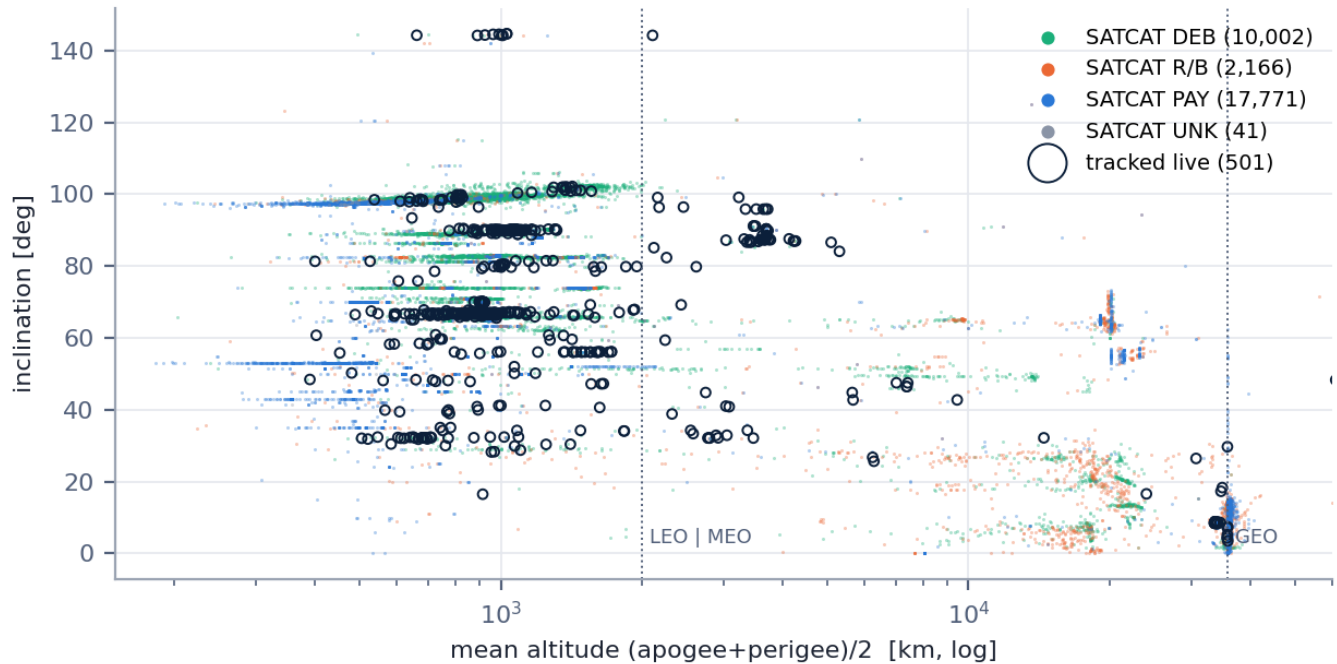
### HOW IT'S COMPUTED

Owner from the CelesTrak SATCAT owner code first, refined by operator name patterns (e.g. US-owned STARLINK-\* → SpaceX). Inclination of live objects from the angular-momentum vector.

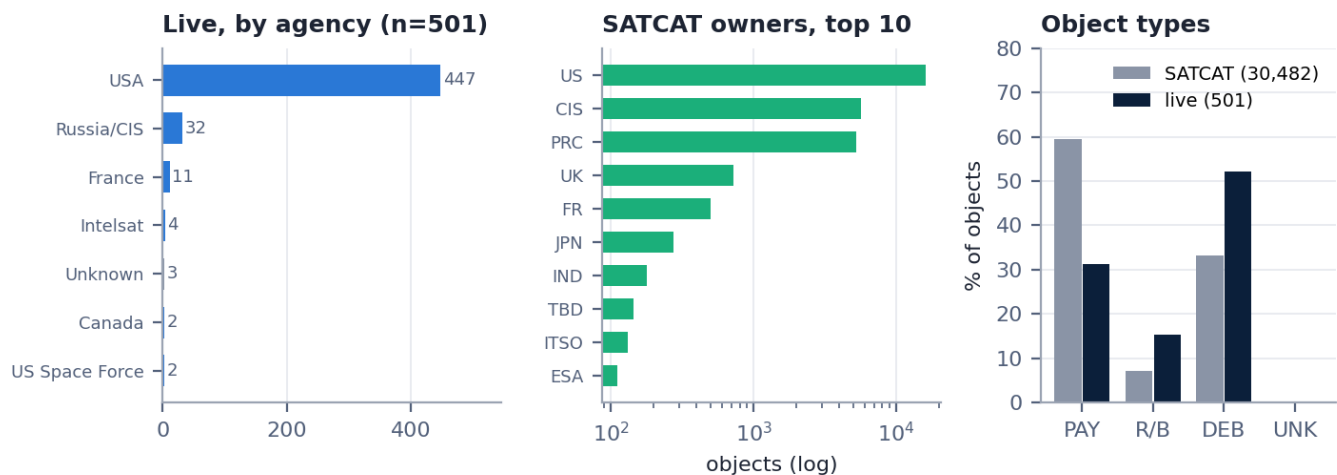
### WHY TRUST IT

Only 3 of 501 live objects (0.6 %) have no SATCAT owner; unknown owners, debris and rocket bodies are never commandable.

### Orbital regimes: full on-orbit catalogue vs the objects tracked live



Source: Grey/colour dots: backend/app/data/satcat\_snapshot.csv (source=<https://celestrak.org/pub/satcat.csv> retrieved=2026-10-08T07:25Z rows=30482 filter=...), non-decayed rows with orbit data. Rings: live states from GET /api/satellites, inclination computed as  $\arccos(h_z/|h|)$  with  $h = r \times v$ .



Source: Left: GET /api/agencies (SATCAT owner + operator-pattern attribution, attribution sources {'satcat\_id': 482, 'satcat\_id+operator\_pattern': 13, 'satcat\_name': 3, 'none': 3}). Middle/right: satcat\_snapshot.csv owner and object\_type columns; live types by NORAD-id lookup.

## 03

## Conjunction screening

a 24-hour future window, not 'distance now'

## WHAT IT IS

Every pair that comes within 25 km in the next 24 h, with its time of closest approach (TCA), miss distance, relative speed and B-plane geometry.

## HOW IT'S COMPUTED

SatrecArray SGP4 grid every 60 s → perigee/apogee shell prefilter → KD-tree query\_pairs per step → linear relative-motion filter → bounded Brent minimisation of the exact SGP4 distance.

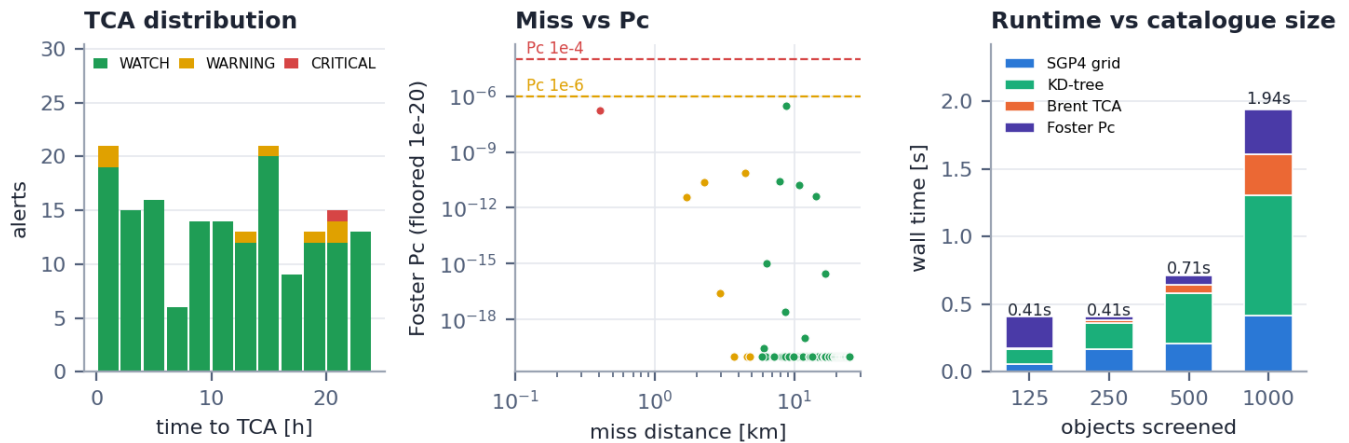
## WHY TRUST IT

tests/test\_screening\_real.py compares against a brute-force 1 s + 1 ms scan (agree to 1 s / 0.1 km) and finds a pair that is > 2000 km apart now but meets in 2 h.

## Closest approach inside a step

$$t^* = -(\Delta r \cdot \Delta v) / |\Delta v|^2, \text{ miss} \approx |\Delta r + \Delta v t^*| \rightarrow \text{Brent refine on } |r_1(t) - r_2(t)|_{\text{SGP4}}$$

Severity: CRITICAL if  $P_c \geq 10^{-4}$  or miss < 1 km; WARNING if  $P_c \geq 10^{-6}$  or miss < 5 km; else WATCH



Source: Left/middle: GET /api/alerts, 170 screened pairs from the live 500-object catalogue. Right: in-process app.core.screening.screen() on the first N TLEs of the bundled snapshot, per-stage timings from its stats dict (this laptop).

Measured: 500 objects → 190 pairs in 0.712 s (KD radius 608.9 km). Runtime grows sub-quadratically because the KD-tree only visits spatial neighbours.

## Top 8 live alerts by Foster Pc (GET /api/alerts)

| Pair                                       | TCA in | Miss      | v_rel      | Foster Pc              | Severity | Decision     | TLE age [d] |
|--------------------------------------------|--------|-----------|------------|------------------------|----------|--------------|-------------|
| THOR ABLESTAR R/B × THOR AGENA D R/B       | 20.7 h | 8.801 km  | 9.34 km/s  | $3.00 \times 10^{-7}$  | WATCH    | MONITOR (12) | 2.3/2.1     |
| THOR ABLESTAR DEB × TITAN 3C TRANSTAGE DEB | 21.1 h | 0.406 km  | 11.02 km/s | $1.72 \times 10^{-7}$  | CRITICAL | MONITOR (26) | 2.0/2.1     |
| SL-8 DEB × DELTA 1 DEB                     | 21.0 h | 4.480 km  | 13.38 km/s | $7.41 \times 10^{-11}$ | WARNING  | NONE (0)     | 2.2/2.1     |
| THOR ABLESTAR DEB × THOR AGENA B R/B       | 5.5 h  | 7.860 km  | 12.40 km/s | $2.80 \times 10^{-11}$ | WATCH    | NONE (11)    | 1.5/1.5     |
| THOR ABLESTAR DEB × THOR ABLESTAR DEB      | 0.7 h  | 2.284 km  | 12.51 km/s | $2.38 \times 10^{-11}$ | WARNING  | NONE (20)    | 1.6/1.4     |
| THOR ABLESTAR DEB × THOR ABLESTAR DEB      | 4.5 h  | 10.822 km | 6.79 km/s  | $1.66 \times 10^{-11}$ | WATCH    | NONE (15)    | 1.4/1.5     |
| THOR ABLESTAR DEB × DIAMANT DEB            | 8.3 h  | 14.298 km | 11.69 km/s | $4.33 \times 10^{-12}$ | WATCH    | NONE (15)    | 1.7/4.8     |
| THOR ABLESTAR DEB × THOR ABLESTAR DEB      | 1.1 h  | 1.693 km  | 14.01 km/s | $3.65 \times 10^{-12}$ | WARNING  | NONE (7)     | 1.4/1.3     |

## 04

## Collision probability

Foster 2-D Pc with a modelled TLE-age covariance — checked four ways

## WHAT IT IS

The probability that the two hard bodies overlap at TCA, given position uncertainty. This number — not the ML model — drives severity and the action.

## HOW IT'S COMPUTED

Project the combined covariance onto the encounter (B-)plane perpendicular to  $\Delta v$  and integrate the 2-D Gaussian over the hard-body disk: 48-point Gauss-Legendre radially  $\times$  96-point trapezoid in angle.

## WHY TRUST IT

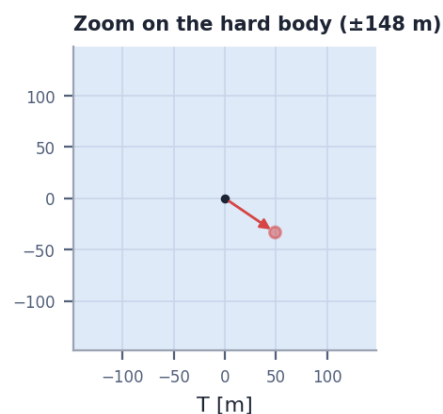
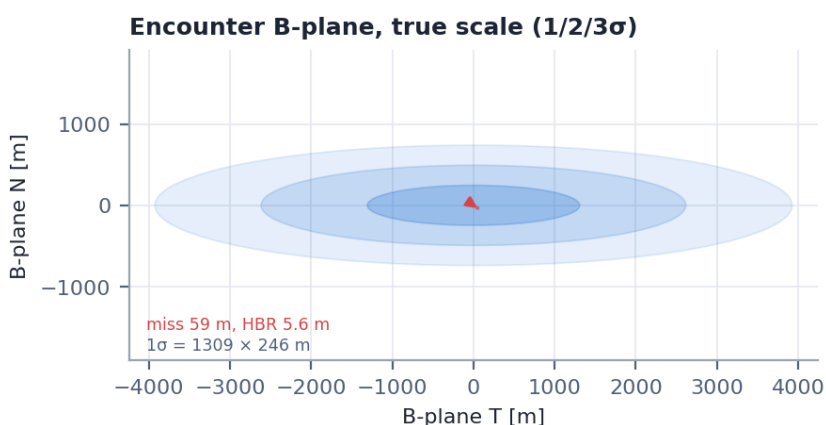
Every alert carries pc\_checks: Chan series (analytic), Monte Carlo (sampling), Alfano max (upper bound). Foster also matches adaptive dblquad to 1e-4 in the test suite.

## Foster (1992) 2-D probability

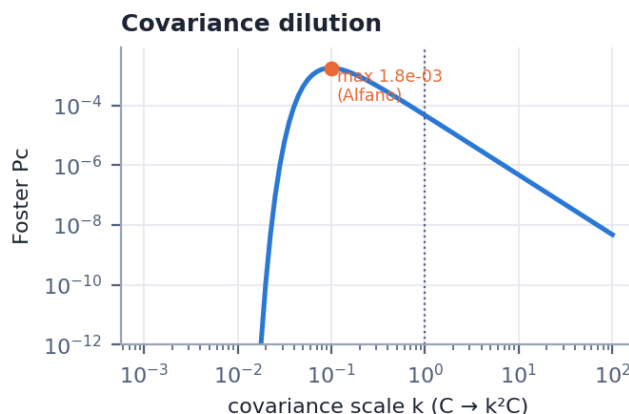
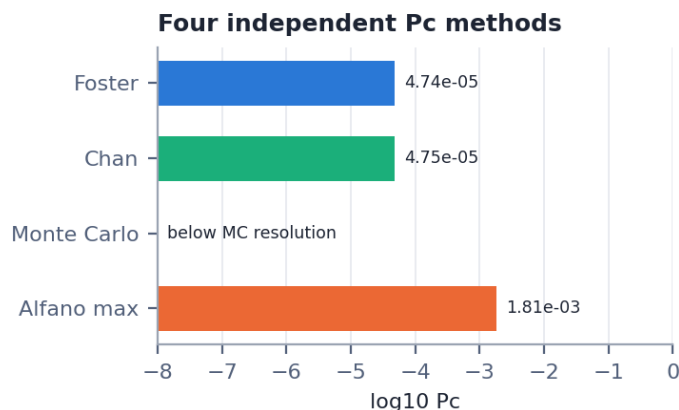
$$P_c = (1 / 2\pi\sqrt{|C_B|}) \iint_{|x| \leq HBR} \exp(-\frac{1}{2} (x-b)^T C_B^{-1} (x-b)) dx$$

$$\sigma_{R,T,N}(\text{age}) = \sigma_0 + g \cdot |\text{TCA} - \text{epoch}|, \sigma_0 = (0.10, 0.50, 0.15) \text{ km}, g = (0.10, 1.00, 0.10) \text{ km/day}$$

HBR = sum of per-object radii (SATCAT RCS class, else type default; ISS 55 m).



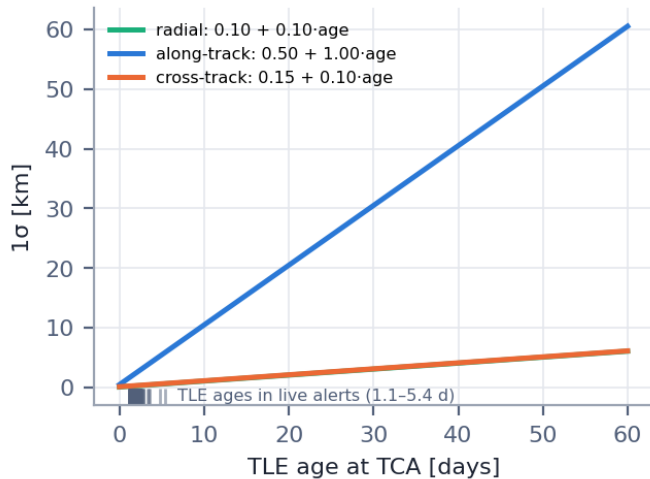
Source: Alert 728-99901 (GGSE 1 (GGRS)  $\times$  SCN CROSSER-B (synthetic)) from GET /api/alerts during the cascade\_demo run: b\_t\_km/b\_n\_km, covariance\_ellipse (a, b, angle at sigma\_level) and hbr\_km.



Source: Left: alert.pc\_checks of 728-99901 (app/core/pc\_methods.py; MC with 200000 samples). Right: app.core.screening.foster\_pc re-evaluated on the same B-plane with the covariance scaled by  $k^2$  (Alfano's maximum is the peak).

For the demo alert, Foster  $4.74 \times 10^{-5}$  vs Chan  $4.75 \times 10^{-5}$ : spread 0.0 decades  $\rightarrow$  **consistent = True**. Monte Carlo (200000 samples, 6 hits) is below its resolution, so it reports null instead of a noisy number. Alfano's maximum 0.00181 is what Pc could reach if the covariance size were wrong.

### TLE-age covariance model



### Why model the covariance?

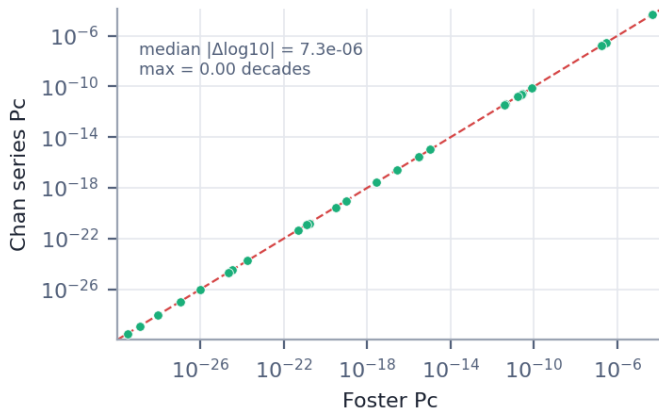
TLEs carry no covariance. We grow a diagonal RTN  $1\sigma$  linearly with TLE age, dominated by the along-track term ( $\sim 1$  km/day). Stale TLEs therefore *dilute*  $P_c$ : with a huge ellipse the probability mass spreads far beyond the hard body. The dilution curve above shows the effect directly — shrinking the covariance first raises  $P_c$ , then lowers it once the miss vector lies outside the ellipse.

Alerts are flagged **stale\_tle** when either TLE is more than 30 days old; short-encounter validity ( $|\Delta v| \geq 0.1$  km/s) is flagged per alert.

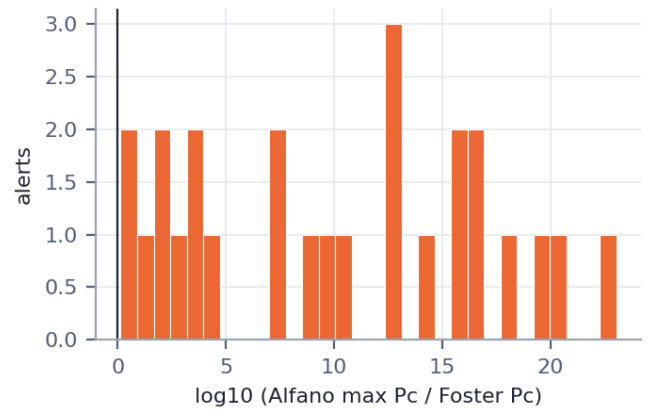
Source: `app.core.screening.tle_age_sigmas_km` ( $\sigma_0$  Flohrer et al. 2008; growth Vallado & Cefola 2012). Rug: `tle_age_days` of the live alerts.

### Do the methods agree on every alert, not just the demo?

#### Foster vs Chan on 26 live alerts



#### Head-room if the covariance size is wrong



Source: `alert.pc_checks` of every screened alert in GET `/api/alerts` (`cascade_demo` run);  $\text{Alfano} \geq \text{Foster}$  by construction (upper bound).

# 05

## ML surrogate & explainability

XGBoost on Foster-labelled encounters — correlation, PCA, SHAP

### WHAT IT IS

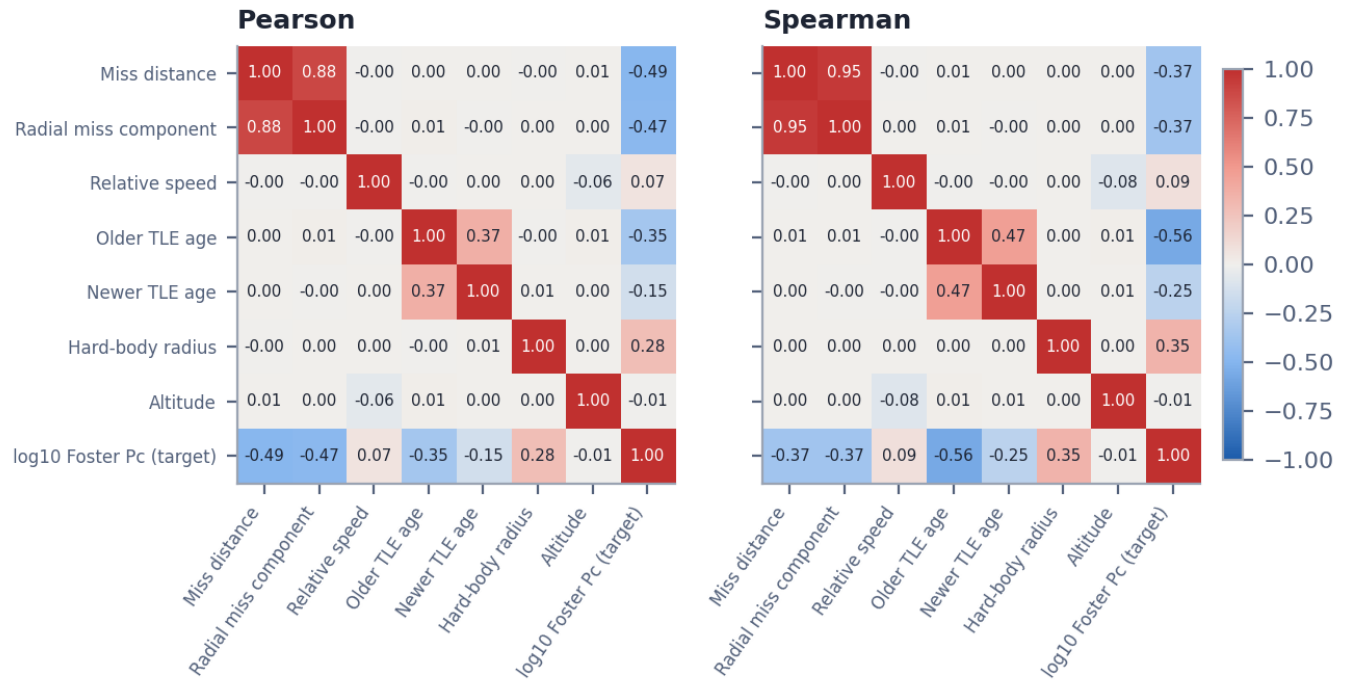
A gradient-boosted regressor of  $\log_{10}$  Pc from 7 encounter features, trained on 42,000 simulated encounters labelled by numerically integrated Foster Pc.

### HOW IT'S COMPUTED

Seeded generator (seed 20261008), 70/10/20 split with early stopping; explainability computed at training time and served by GET /api/analytics/model.

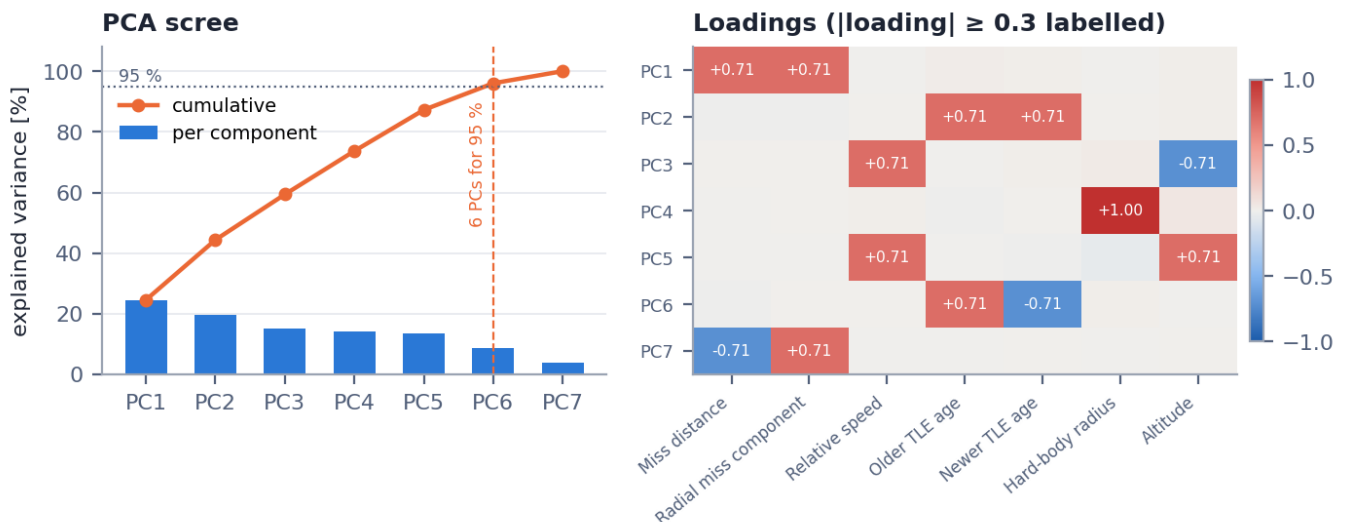
### WHY TRUST IT

It beats a miss-distance-only baseline by an order of magnitude, is evaluated on a held-out split, and is only advisory: it never writes probability\_of\_collision.



Source: GET /api/analytics/model → correlation (pairwise deletion (each coefficient uses rows where both variables are present); 42000 training rows). Last row/column = target log10 Pc.

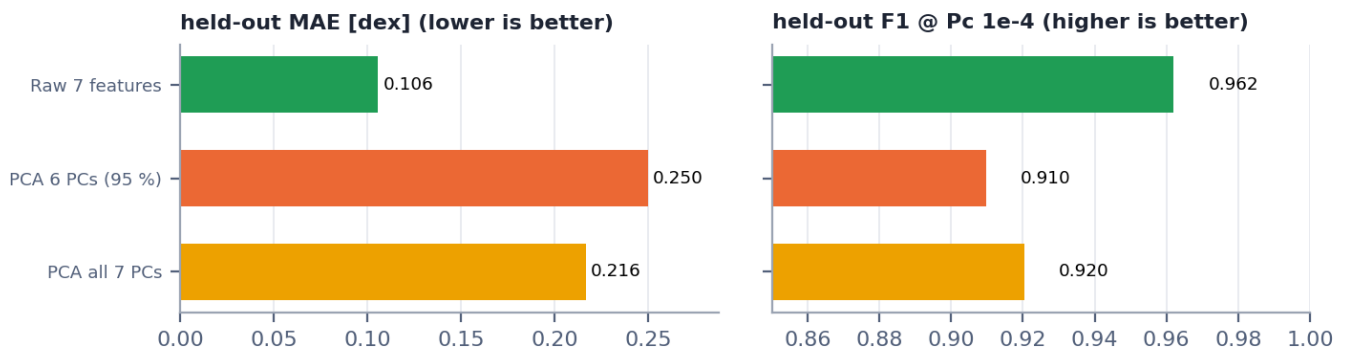
**Reading it:** Miss distance ~ Radial miss component  $\rho = 0.95$ ; Older TLE age ~ Newer TLE age  $\rho = 0.47$ . Strongest link to the target: Older TLE age (Spearman -0.56) — older TLEs mean bigger ellipses and lower Pc.



Source: GET /api/analytics/model → pca (unit eigenvectors of the feature correlation matrix (rows = components); training-split median per feature, PCA only (XGBoost uses native missing-value branches)).

- ▶ PC1 (24.6% of variance): +0.71 Radial miss component, +0.71 Miss distance -- these move together.
- ▶ PC2 (19.8% of variance): +0.71 Newer TLE age, +0.71 Older TLE age -- these move together.
- ▶ PC3 (15.0% of variance): +0.71 Relative speed, -0.71 Altitude -- a contrast: the + and - groups move in opposite directions.

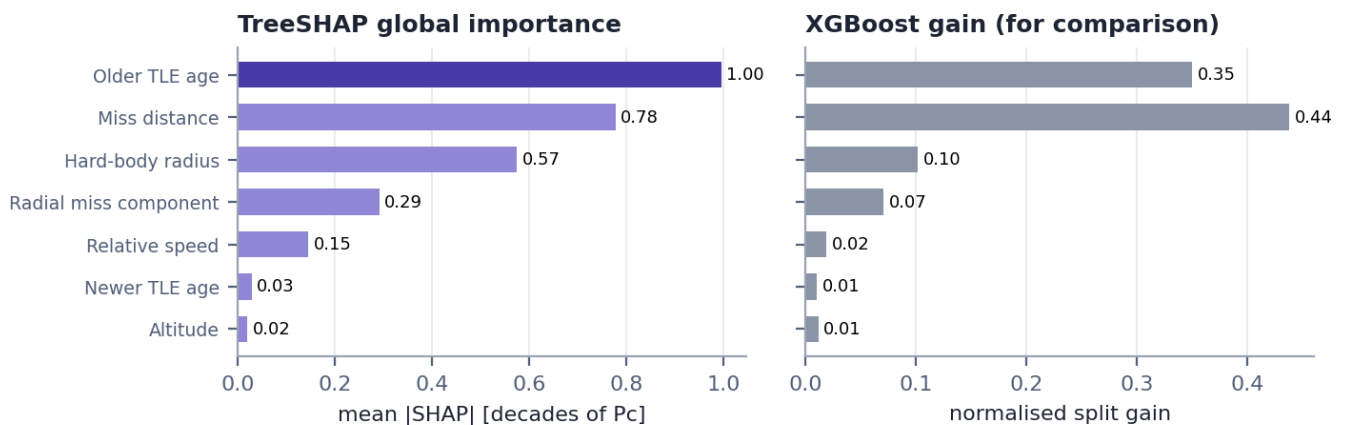




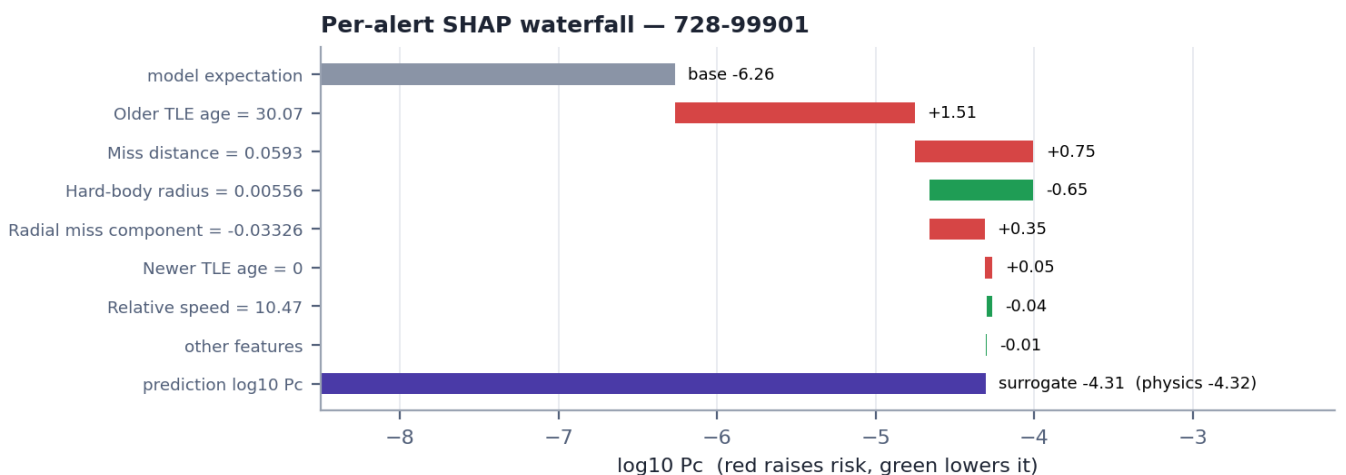
Source: GET /api/analytics/model → pca\_vs\_raw: same rows, split, seed and XGBoost hyper-parameters; PCA fitted on training rows only.

#### Verdict on PCA (measured, not assumed)

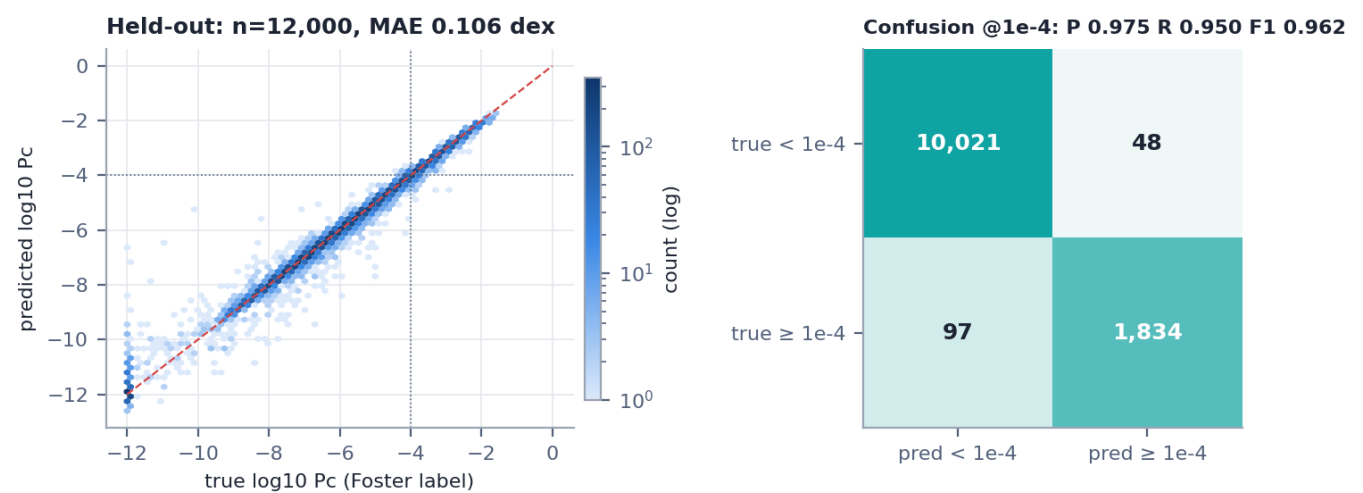
Raw features win: XGBoost on the 7 raw features scores held-out MAE 0.106 dex / F1@1e-4 0.962, versus 0.250 dex / 0.910 on the first 6 principal components (95% variance). Even with all 7 components (a pure rotation, no information lost) MAE is 0.216 dex: trees split one axis at a time, so rotating physically meaningful axes (miss distance, TLE age, HBR) into mixtures makes the Pc boundary harder to carve, and the dropped low-variance components still carry Pc signal. PCA is used here to describe the inputs, not as a preprocessing step.



Source: GET /api/analytics/model → shap\_global (exact path-dependent TreeSHAP (xgboost pred\_contribs=True) on held-out rows, 2000 held-out rows, additivity error 6.68e-06) and split\_importance.gain.



Source: alert.ml of 728-99901 from GET /api/alerts (treeshap\_exact); base\_log10 +  $\sum$  contributions = log10 surrogate Pc.



Source: Held-out split regenerated with `app.ml.train_risk_surrogate.generate_encounters` (seed 20261008, n=60,000, 70/10/20) and scored with the saved `risk_model_xgb.json` (6 s). Confusion counts match the model card.

Model card

|                          |                                                                                                                      |                        |                                                                                                 |
|--------------------------|----------------------------------------------------------------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------|
| Model                    | xgboost-pc-surrogate v1                                                                                              | Task                   | Pc surrogate (log10 Foster Pc regression)                                                       |
| Train / val / held-out   | 42,000 / 6,000 / 12,000                                                                                              | Boosting rounds        | 88                                                                                              |
| Held-out MAE / RMSE      | 0.1056 / 0.2409 dex                                                                                                  | MAE with masked inputs | 0.3372 dex                                                                                      |
| P / R / F1 @ 1e-4        | 0.9745 / 0.9498 / 0.962                                                                                              | P / R / F1 @ 1e-6      | 0.9856 / 0.9788 / 0.9822                                                                        |
| Baseline (miss only) MAE | 1.5654 dex                                                                                                           | Constant baseline MAE  | 1.7389 dex                                                                                      |
| Features                 | miss_distance_km, radial_miss_km, relative_speed_kms, tle_age_max_h, tle_age_min_h, hbr_km, altitude_km              | Not seen by model      | combined covariance, B-plane projected sigmas, B-plane miss direction (only  radial  component) |
| Label                    | Foster Pc on simulated encounters (numerical integration of the combined TLE-age covariance over the hard-body disk) | Trained                | 2026-10-08T08:33 · xgboost 3.4.1                                                                |

**Honesty note:** the separate trajectory RNN (numpy tanh RNN (12 hidden units, 8-step history)) scores MAE 459.164 km vs 20.513 km for linear extrapolation; it is marked **experimental** and feeds nothing in the alert path.

## 06

## Decision score

one well-defined number across physics, ML, cascade and manoeuvre

## WHAT IT IS

A 0–100 score and an action (MANOEUVRE / PREPARE / MONITOR / NONE) with a confidence level and a generated one-sentence rationale on every alert.

## HOW IT'S COMPUTED

Weighted sum of four normalised components; the action comes from the physics  $P_c$  alone, and ML + cascade can only escalate PREPARE → MANOEUVRE when both agree.

## WHY TRUST IT

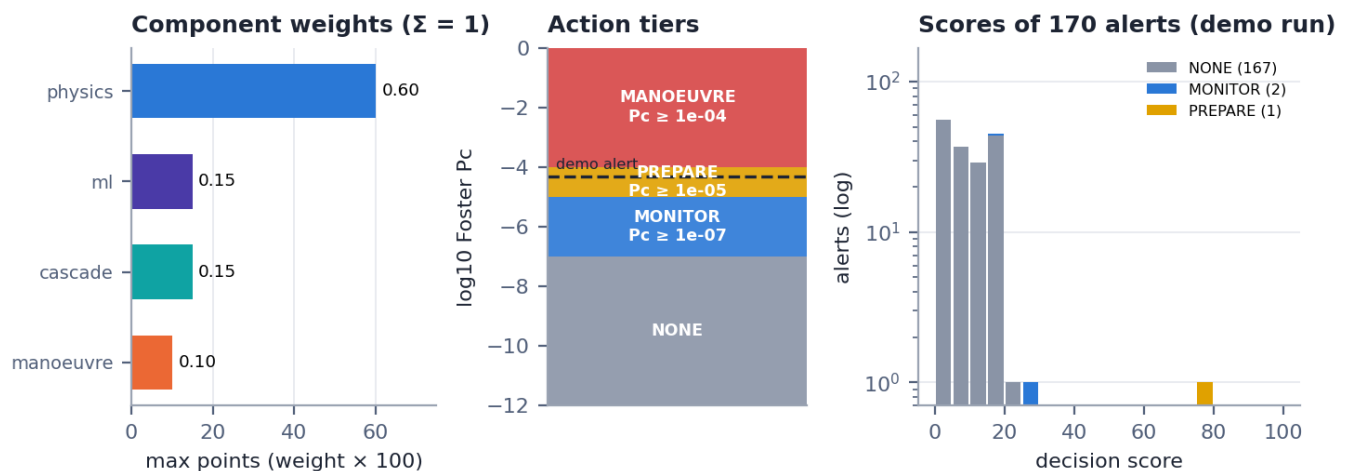
Unit-tested: weights sum to 1, physics outweighs everything else combined, score monotone in  $P_c$ , points add up to the score, thresholds follow NASA CARA practice.

## Decision score (app/core/decision.py)

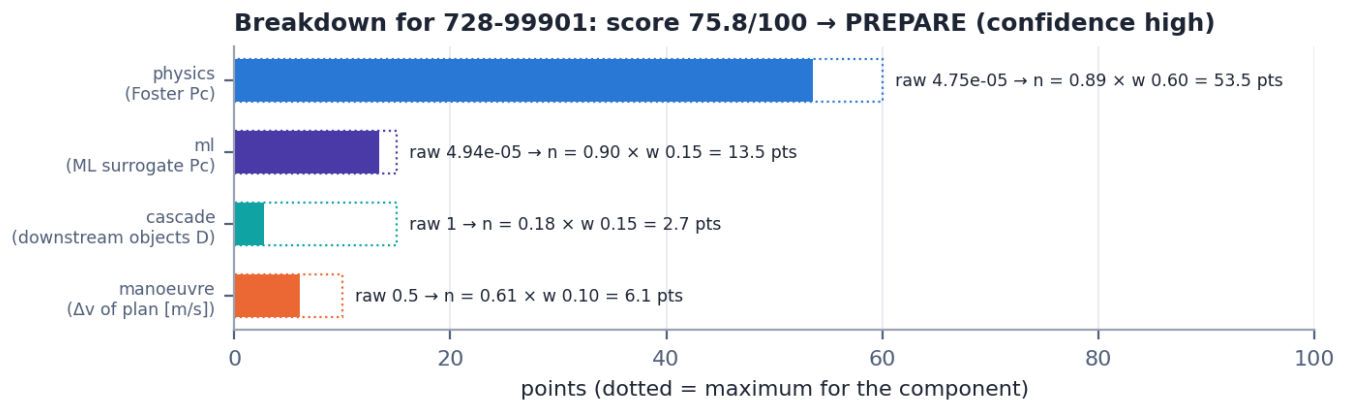
$$L(p) = \text{clip}((\log_{10} p + 7) / 3, 0, 1) - 0 \text{ at } 10^{-7}, 1 \text{ at } 10^{-4}$$

$$\text{score} = 100 \cdot [0.60 L(P_{c_{\text{Foster}}}) + 0.15 L(P_{c_{\text{ML}}}) + 0.15 (1 - e^{-D/5}) + 0.10 e^{-\Delta v/1 \text{ m/s}} \cdot s_{\text{safe}}]$$

$D$  = downstream objects in the cascade graph;  $s_{\text{safe}} = 1$  (cascade-safe), 0.5 (not), 0 (no plan)



Source: app.core.decision.WEIGHTS /  $P_c$  \* constants; score histogram from alert.decision of every alert in GET /api/alerts while the cascade\_demo scenario was loaded.



Source: alert.decision.components of 728-99901 (GET /api/alerts, cascade\_demo run).

**Rationale (generated):** Foster  $P_c$   $4.75e-05$  is in the PREPARE band [ $1e-05$ ,  $1e-04$ ]; ML surrogate  $4.94e-05$  (0.02 decades from physics); Foster/Chan spread 0.00 decades; 1 downstream object; verified manoeuvre 0.500 m/s → PREPARE, confidence high, score 76/100.

## 07

## Cascade-safe avoidance

re-propagated burns, secondary-conjunction check, 12 engines

## WHAT IT IS

For each risky alert: a ranked set of burns ( $\pm R/\pm S/\pm W \times 0.01\text{--}2$  m/s), each with its new Pc, secondary conjunctions, propellant and per-engine feasibility.

## HOW IT'S COMPUTED

Each candidate is applied to a copy of the trajectory, the encounter re-screened and Pc recomputed with the production `compute_pc`; shortlisted options are re-screened for 24 h against the whole catalogue + live debris.

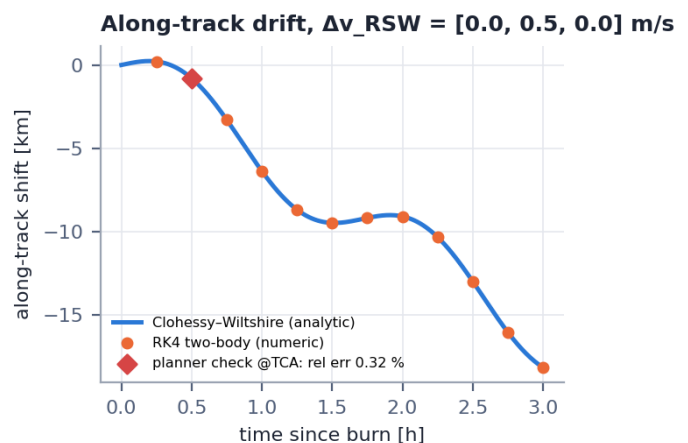
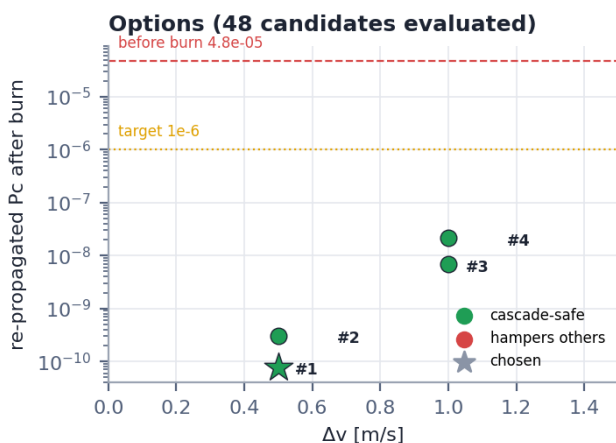
## WHY TRUST IT

The chosen burn is the smallest  $\Delta v$  that is cascade-safe; tests show the naive min- $\Delta v$  burn can hit a third object and is rejected. CW analytic drift and RK4 mass integration verify each option.

## Checks attached to every option

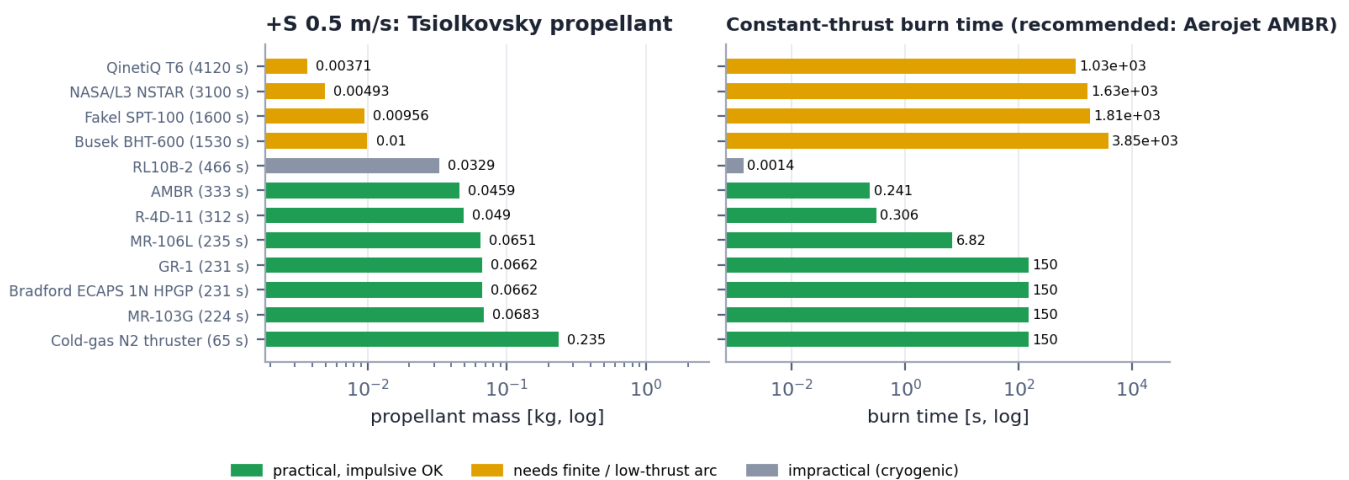
Clohessy-Wiltshire:  $y(t) = (2\dot{x}_0/n)(\cos nt - 1) + (\dot{y}_0/n)(4 \sin nt - 3nt)$

Tsiolkovsky:  $m_p = m_0(1 - e^{-\Delta v/(I_{sp} g_0)})$ ;  $\text{finite\_burn\_ok} \Leftrightarrow t_{\text{burn}} \leq 0.1 \cdot \text{lead time and } \leq 1/10 \text{ orbit}$



Source: `recommended_maneuver.options` of 728-99901 (`GET /api/alerts, cascade_demo` run; numbers # match the table below). Right: `app.core.analytic_checks cw_position` vs `numeric_along_track_shift_km` on a circular orbit with the option's mean motion; red diamond = the planner's own analytic check.

| #    | Burn       | $\Delta v$ [m/s] | New miss [km] | New Pc                 | Secondaries (24 h) | Cascade-safe | Prop. [kg] | Engine       |
|------|------------|------------------|---------------|------------------------|--------------------|--------------|------------|--------------|
| #1 ★ | +S 0.5 m/s | 0.5              | 1.409         | $7.70 \times 10^{-11}$ | 0 (max Pc 0)       | yes          | 0.070      | Aerojet AMBR |
| #2   | -S 0.5 m/s | 0.5              | 1.308         | $3.07 \times 10^{-10}$ | 0 (max Pc 0)       | yes          | 0.070      | Aerojet AMBR |
| #3   | +R 1 m/s   | 1                | 2.048         | $6.85 \times 10^{-9}$  | 0 (max Pc 0)       | yes          | 0.139      | Aerojet AMBR |
| #4   | -R 1 m/s   | 1                | 1.929         | $2.15 \times 10^{-8}$  | 0 (max Pc 0)       | yes          | 0.139      | Aerojet AMBR |



Source: `options[0].engines` of the demo alert (`app.core.propulsion.evaluate_engines`; mass 300.0 kg [`satcat_rcs_class:MEDIUM`]); `finite_burn_ok` = burn  $\leq 10\%$  of lead time and  $\leq 1/10$  orbit.

Selection rule: *min*  $\Delta v$  among cascade-safe options achieving  $P_c < 1e-06$  · mover mass 300.0 kg (`satcat_rcs_class:MEDIUM`),  $I_{sp}$  220.0 s (default\_hydrazine).

## 08

## Collision → debris → cascade

NASA Standard Breakup Model, propagated fragments, graph to agencies

## WHAT IT IS

What happens if the demo pair collides:  
GGSE 1 (GGRS) (500 kg,  
satcat\_rcs\_medium) × SCN CROSSER-B  
(synthetic) (800 kg) at 10.47 km/s.

## HOW IT'S COMPUTED

Both objects propagated to the predicted  
TCA; SBM samples size, area-to-mass and  
ejection  $\Delta v$  per fragment (seeded);  
fragments are integrated with J2 + drag  
and screened against the catalogue for 6  
h.

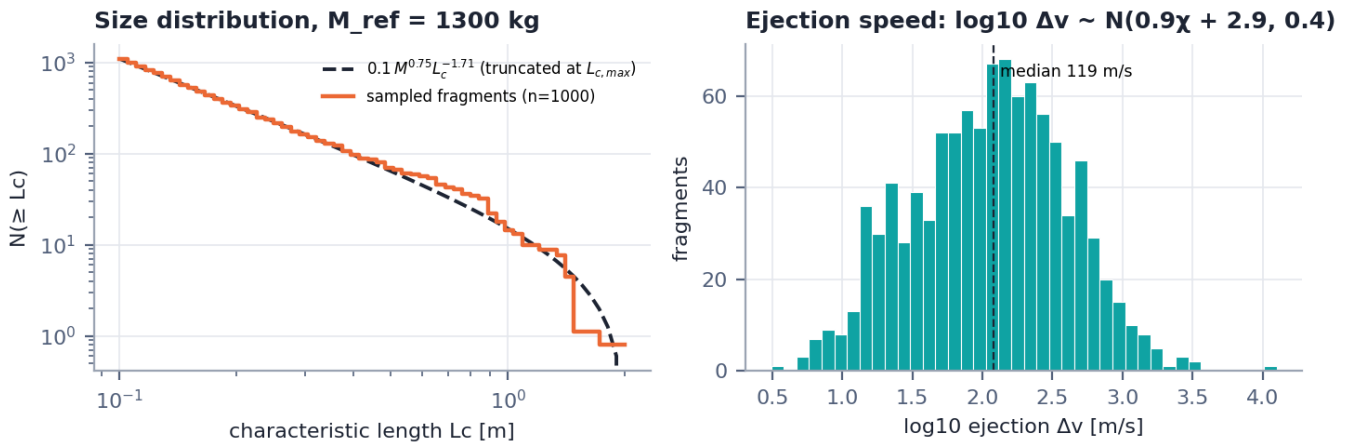
## WHY TRUST IT

Fragment count equals  $0.1 \cdot M^{0.75} \cdot L_c^{-1.71}$   
(live check); momentum is re-centred,  $\Delta v$   
capped below escape, energy drift  $< 1e-5$   
without drag (tests/test\_debris\_real.py).

## NASA SBM (Johnson et al. 2001)

$EMR = \frac{1}{2} m_p v^2 / M_t = 34280 \text{ J/g} \geq 40 \text{ J/g} \rightarrow \text{catastrophic}; M = m_1 + m_2 = 1300 \text{ kg}$

$N(\geq 10 \text{ cm}) = 0.1 \cdot M^{0.75} \cdot 0.1^{-1.71} \rightarrow 1104 \text{ fragments}; \log_{10} \Delta v \sim N(0.9\chi + 2.9, 0.4), \chi = \log_{10}(A/M)$

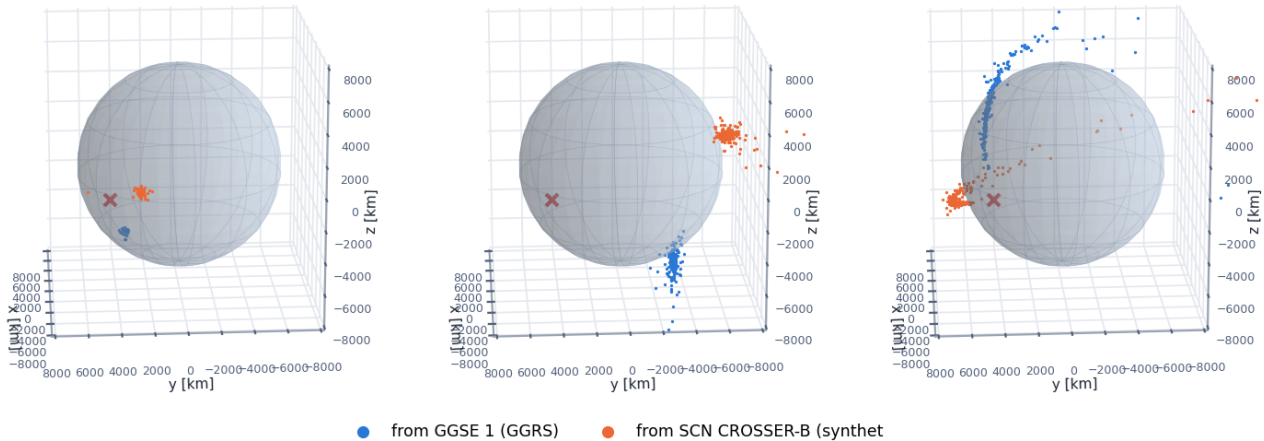


Source: `app.core.breakup.simulate_breakup` re-run with the event's own TCA states, masses and seed 3355768798 (event evt-728-99901-20261008T100835Z);  $n_{\text{total}}$  1104 vs event 1104,  $\Delta v$  median identical to the backend event stats.

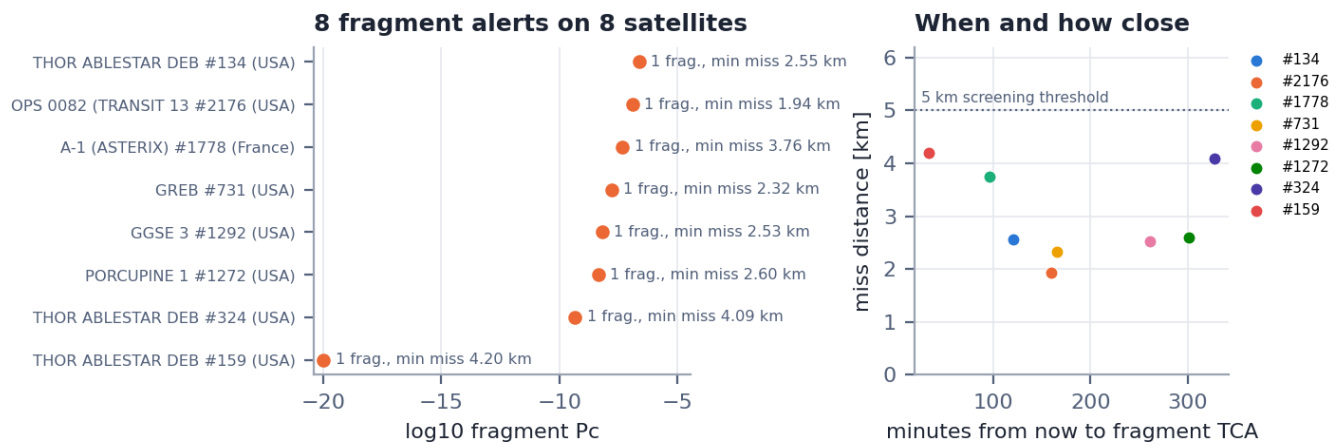
+5 min: 999 fragments &gt; 100 km

+30 min: 955 fragments &gt; 100 km

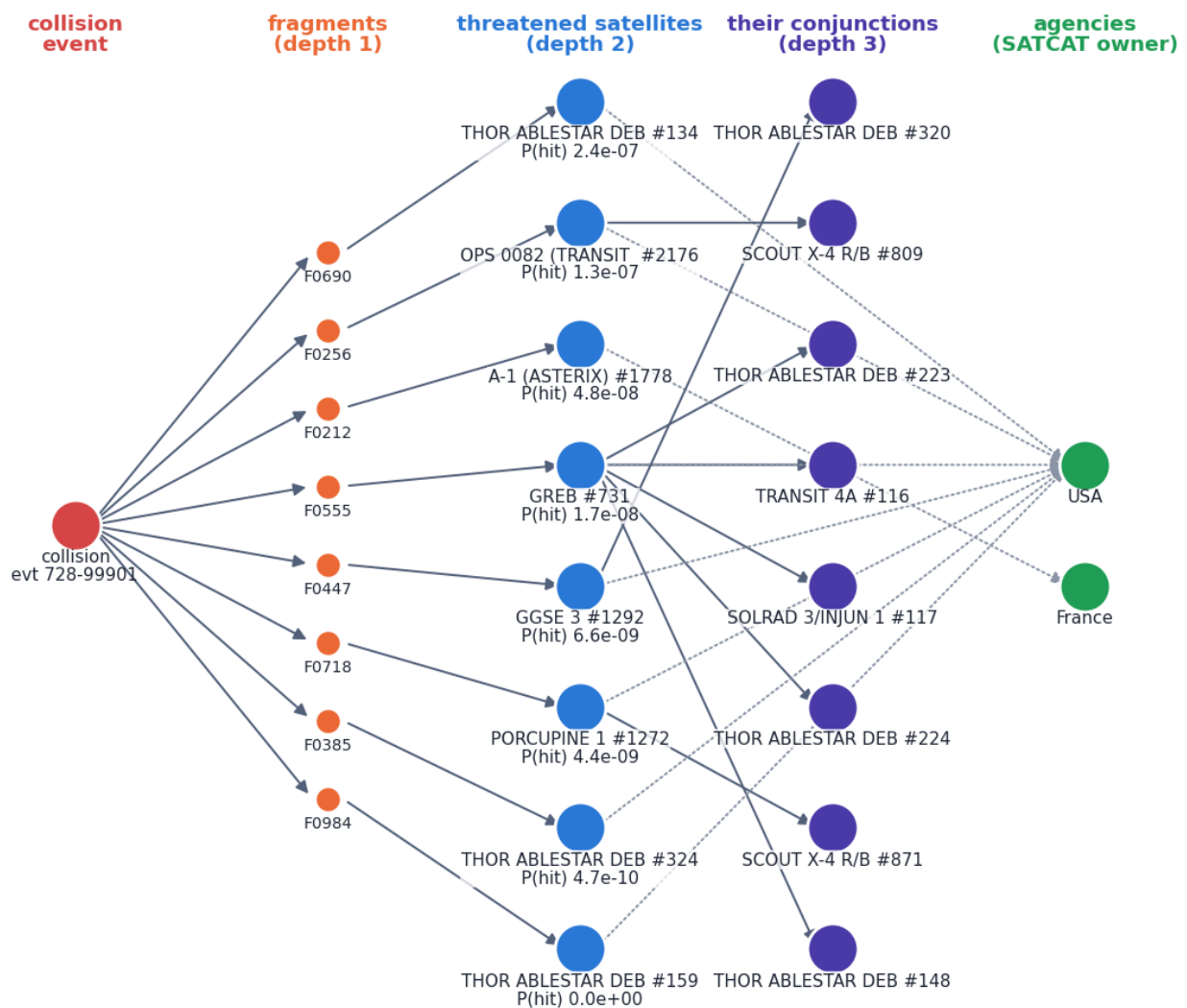
+90 min: 921 fragments &gt; 100 km



Source: Fragments from the reproduced breakup propagated with `app.core.breakup.propagate` (RK4, two-body + J2 + Vallado drag,  $Cd \cdot A/M$ ), shown in 3-D ECI/TEME at true scale (equal axes, Earth sphere  $R = 6378 \text{ km}$ ) for both parents; camera placed above the impact point (x). Fragments below 100 km are removed.



Source: source=='debris' alerts from GET /api/alerts after POST /api/debris/simulate and a pipeline recompute (6 h window, 30 s step, 5 km threshold, isotropic Foster Pc); agency = SATCAT owner.



Source: Graph built from GET /api/alerts after the debris recompute: debris alerts give event → fragment → satellite edges, screening alerts give the satellites' own conjunctions (top 8 by Pc), agency from alert.sat1.agency; P(hit) from node\_probabilities.

Debris screening: 1000 fragments × 501 satellites, 6 h at 30 s steps → 8 alerts in 101.6 s. Cascade depth is the BFS hop count from the collision event; P(hit) combines all incident alerts assuming independence. This is a graph over predicted encounters in the screening window, not a multi-year population model.

| 09 Physics cross-validation                                                                                                                                                                                         |  |                                                                                                                                                                                                                          |                                          |                                                                                                                                                                                                    |                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| GET /api/physics/validation — 18/18 checks pass (computed live)                                                                                                                                                     |  |                                                                                                                                                                                                                          |                                          |                                                                                                                                                                                                    |                                                           |
| <b>WHAT IT IS</b><br>A battery of standard-formula checks that runs inside the live backend (cached 60 s), so judges can hit the endpoint and see the same table.                                                   |  | <b>HOW IT'S COMPUTED</b><br>Each check recomputes a quantity with an independent formulation (closed form, series, sampling or numerical integration) and compares it to our production code against a stated tolerance. |                                          | <b>WHY TRUST IT</b><br>Methods come from the literature (Foster 1992, Chan 2008, Alfano 2005, Johnson 2001, Vallado); the tolerance kind (relative / absolute / inequality) is reported per check. |                                                           |
| Check (standard formula)                                                                                                                                                                                            |  | Ours                                                                                                                                                                                                                     | Reference                                | Error                                                                                                                                                                                              | Tolerance                                                 |
| <b>Pc Foster vs Chan series — G1 isotropic</b><br>$P_c = e^{-\{v/2\} \sum (v/2)^m/m! [1 - e^{-\{u/2\} \sum_{k \leq m} (u/2)^k/k!}]}$ , $u=R^2/\sigma x \sigma y$ , $v=x^2/\sigma x^2+y^2/\sigma y^2$                |  | 0.00376742                                                                                                                                                                                                               | 0.00376742                               | rel $4.60 \times 10^{-16}$                                                                                                                                                                         | $\leq 0.02$ <b>PASS</b>                                   |
| <b>Pc Foster vs Monte Carlo (N=200000) — G1 isotropic</b><br>$P_c \approx (1/N) \sum 1[ x_i  \leq HBR]$ , $x_i \sim N(b, C)$ ; tolerance $4\sigma_{MC} = 4 \sqrt{p(1-p)/N}$                                         |  | 0.00376742                                                                                                                                                                                                               | 0.00382                                  | abs $5.26 \times 10^{-5}$                                                                                                                                                                          | $\leq 5.52 \times 10^{-4}$ <b>PASS</b>                    |
| <b>Pc Foster vs Chan series — G2 anisotropic 5:1</b><br>$P_c = e^{-\{v/2\} \sum (v/2)^m/m! [1 - e^{-\{u/2\} \sum_{k \leq m} (u/2)^k/k!}]}$ , $u=R^2/\sigma x \sigma y$ , $v=x^2/\sigma x^2+y^2/\sigma y^2$          |  | 0.00182892                                                                                                                                                                                                               | 0.00183132                               | rel 0.00131                                                                                                                                                                                        | $\leq 0.02$ <b>PASS</b>                                   |
| <b>Pc Foster vs Monte Carlo (N=200000) — G2 anisotropic 5:1</b><br>$P_c \approx (1/N) \sum 1[ x_i  \leq HBR]$ , $x_i \sim N(b, C)$ ; tolerance $4\sigma_{MC} = 4 \sqrt{p(1-p)/N}$                                   |  | 0.00182892                                                                                                                                                                                                               | 0.001725                                 | abs $1.04 \times 10^{-4}$                                                                                                                                                                          | $\leq 3.71 \times 10^{-4}$ <b>PASS</b>                    |
| <b>Pc Foster vs Chan series — G3 rotated 3.3:1, large HBR</b><br>$P_c = e^{-\{v/2\} \sum (v/2)^m/m! [1 - e^{-\{u/2\} \sum_{k \leq m} (u/2)^k/k!}]}$ , $u=R^2/\sigma x \sigma y$ , $v=x^2/\sigma x^2+y^2/\sigma y^2$ |  | 0.00230308                                                                                                                                                                                                               | 0.00230057                               | rel 0.00109                                                                                                                                                                                        | $\leq 0.02$ <b>PASS</b>                                   |
| <b>Pc Foster vs Monte Carlo (N=200000) — G3 rotated 3.3:1, large HBR</b><br>$P_c \approx (1/N) \sum 1[ x_i  \leq HBR]$ , $x_i \sim N(b, C)$ ; tolerance $4\sigma_{MC} = 4 \sqrt{p(1-p)/N}$                          |  | 0.00230308                                                                                                                                                                                                               | 0.00223                                  | abs $7.31 \times 10^{-5}$                                                                                                                                                                          | $\leq 4.22 \times 10^{-4}$ <b>PASS</b>                    |
| <b>Pc Foster vs exact Rice distribution (isotropic)</b><br>$P_c = F_{\chi^2}(R^2/\sigma^2; k=2, \lambda= b ^2/\sigma^2)$                                                                                            |  | 0.00376742                                                                                                                                                                                                               | 0.00376742                               | rel $2.30 \times 10^{-16}$                                                                                                                                                                         | $\leq 1.00 \times 10^{-6}$ <b>PASS</b>                    |
| <b>Alfano max-Pc: closed form vs numerical covariance-scale maximisation</b><br>$P_{c\_max} = HBR^2 / (e \cdot \sigma x \sigma y \cdot b^T C^{-1} b)$                                                               |  | 0.00654017                                                                                                                                                                                                               | 0.00654008                               | rel $1.33 \times 10^{-5}$                                                                                                                                                                          | $\leq 0.01$ <b>PASS</b>                                   |
| <b>Alfano max-Pc is an upper bound of Foster Pc — G3</b><br>$P_{c\_max} \geq P_c(k=1)$                                                                                                                              |  | 0.00259778                                                                                                                                                                                                               | 0.00230308                               | $\Delta 2.95 \times 10^{-4}$                                                                                                                                                                       | inequality (our_value $\geq$ reference_value) <b>PASS</b> |
| <b>Energy conservation of production J2-RK4 propagator (3 h, ISS orbit)</b><br>$\epsilon = v^2/2 - \mu/r + (\mu J^2 R e^2 / 2 r^3) (3 z^2 / r^2 - 1) = \text{const}$                                                |  | -29.6095 km <sup>2</sup> /s <sup>2</sup>                                                                                                                                                                                 | -29.6095 km <sup>2</sup> /s <sup>2</sup> | rel $6.71 \times 10^{-11}$                                                                                                                                                                         | $\leq 1.00 \times 10^{-8}$ <b>PASS</b>                    |
| <b>Vis-viva semi-major axis of SGP4 arc vs SGP4 mean elements</b><br>$a = 1 / (2/r - v^2/\mu)$ , averaged over one revolution                                                                                       |  | 6731.45 km                                                                                                                                                                                                               | 6731.47 km                               | rel $2.53 \times 10^{-6}$                                                                                                                                                                          | $\leq 0.001$ <b>PASS</b>                                  |
| <b>SGP4 vs J2-RK4 short-arc agreement (30 min)</b><br>$ r_{SGP4}(t) - r_{J2RK4}(t) $ from the same initial state                                                                                                    |  | 0.0395316 km                                                                                                                                                                                                             | 0 km                                     | abs 0.0395                                                                                                                                                                                         | $\leq 1$ <b>PASS</b>                                      |
| <b>CW along-track drift vs numeric (0.1 m/s +S, 3 h, 500 km)</b><br>$y(t) = (\dot{y}_0/n)(4 \sin nt - 3nt)$                                                                                                         |  | -3.44012 km                                                                                                                                                                                                              | -3.44796 km                              | rel 0.00227                                                                                                                                                                                        | $\leq 0.03$ <b>PASS</b>                                   |
| <b>Hohmann <math>\Delta v</math> 500→520 km vs numerical altitude raise</b><br>$\Delta v_1 = \sqrt{\mu/(r_1)}(\sqrt{2r_2/(r_1+r_2)}) - 1$                                                                           |  | 5.52388 m/s                                                                                                                                                                                                              | 5.52388 m/s                              | rel $1.64 \times 10^{-7}$                                                                                                                                                                          | $\leq 0.001$ <b>PASS</b>                                  |
| <b>Vis-viva energy conservation (3 orbits, e=0.01)</b><br>$\epsilon = v^2/2 - \mu/r = -\mu/(2a)$                                                                                                                    |  | -27.8756 km <sup>2</sup> /s <sup>2</sup>                                                                                                                                                                                 | -27.8756 km <sup>2</sup> /s <sup>2</sup> | rel $6.52 \times 10^{-11}$                                                                                                                                                                         | $\leq 1.00 \times 10^{-8}$ <b>PASS</b>                    |
| <b>Rocket equation vs numeric mass integration (1.0 m/s, Isp 220 s)</b><br>$m_p = m_0 (1 - \exp(-\Delta v/(Isp \cdot g_0)))$                                                                                        |  | 0.2317 kg                                                                                                                                                                                                                | 0.2317 kg                                | rel $1.37 \times 10^{-9}$                                                                                                                                                                          | $\leq 1.00 \times 10^{-6}$ <b>PASS</b>                    |
| <b>NASA SBM fragment count <math>\geq 10</math> cm (800 kg + 15 kg @ 10 km/s)</b><br>$N(\geq L_c) = 0.1 M^{0.75} L_c^{-1.71}$ ; catastrophic if $EMR \geq 40 J/g \rightarrow M = m_1 + m_2$                         |  | 778 fragments                                                                                                                                                                                                            | 777.628 fragments                        | abs 0.372                                                                                                                                                                                          | $\leq 1$ <b>PASS</b>                                      |
| <b>SBM sampled size distribution: fraction <math>\geq 0.5</math> m (n=778)</b><br>$P(L_c \geq x) = (x^{-1.71} - L_{c\_max}^{-1.71})/(L_{c\_min}^{-1.71} - L_{c\_max}^{-1.71})$                                      |  | 0.059126                                                                                                                                                                                                                 | 0.0581781                                | abs $9.48 \times 10^{-4}$                                                                                                                                                                          | $\leq 0.0336$ <b>PASS</b>                                 |



10

Independent verification

a separate agent re-derived results outside the production code path

149

independent checks

149/149

pass (after fixes)

3

genuine defects found & fixed

14

areas covered

WHAT IT IS

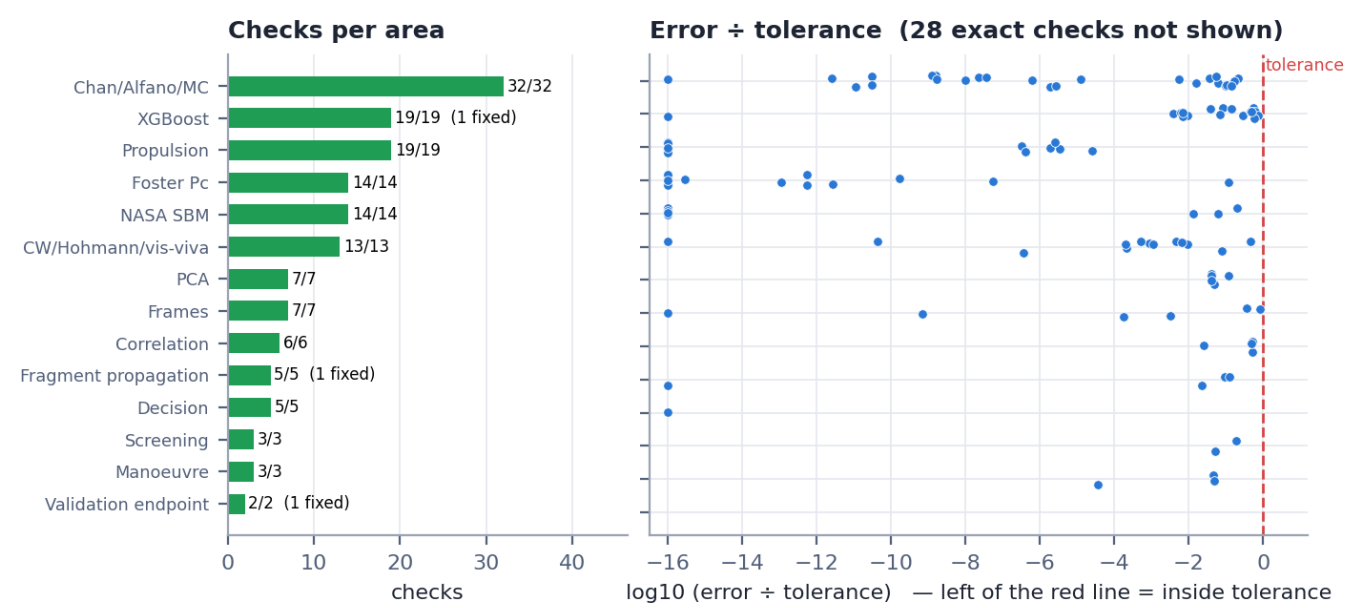
A separate agent recomputed every mathematical component with a different implementation, without reusing the project's code path.

HOW IT'S COMPUTED

scipy dblquad / ncx2 / solve\_ivp (DOP853), sklearn PCA & metrics, raw sgp4 brute-force scans, KS tests, an own re-implementation of the decision formula; each record logs ours vs independent, error and tolerance.

WHY TRUST IT

It found real bugs (below) and they were fixed and re-tested — evidence that the checks have teeth, not just that they pass.



Source: docs/report/verification.json (independent re-implementations: scipy, sklearn, raw sgp4, DOP853 integrators). error = abs\_err or rel\_err, whichever the tolerance applies to; checks with tolerance 0 (exact equality / inequality) are counted on the left only.

Defects found and fixed

| id        | Defect                                                                                                                                                                                                              | Impact                                                                                                                   | Fix                                                                                                                                      |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| PROP-2b   | debris_model._satellite_tracks propagated satellites without a TLE (two-body + J2 RK4) with 60 s substeps. The error was 207 m after 6 h against DOP853.                                                            | About 1σ of the 0.2 km satellite sigma used in the debris Pc, so a real Pc error for scenario / state-vector satellites. | Substep is now SAT_TRACK_MAX_STEP_S = 20 s, which gives 0.45 m on the production 30 s grid. The test is tests/test_verification_real.py. |
| ML-CARD-1 | The model-card text recipe.covariance described an obsolete covariance (σ=σn=0.05 km, σt=1⋅(1+age_h/24) km). The labels actually use the live TLE-age model σ = (0.1, 0.5, 0.15) km + (0.1, 1.0, 0.1) km/day ⋅ age. | Documentation only. Labels and metrics were already correct (re-verified bit-for-bit).                                   | The text is now generated from the screening constants (_covariance_recipe_text), and the card JSON was corrected. No retrain.           |
| VAL-2     | /api/physics/validation reported rel_error = 3.95e+298 for the zero-reference check "SGP4 vs J2-RK4". The cause was abs/1e-300.                                                                                     | A nonsensical number in the API.                                                                                         | rel_error = null when the reference is 0, and pass/fail falls back to the absolute error (routers/physics.py, core/analytic_checks.py).  |



## What was verified — independent method and worst error per area

| Area                    | Independent method                                                                                                       | Worst error                                                                                            |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| PCA                     | Regenerated the 60 000-row set (seed 20261008, same split), then sklearn SimpleImputer(median) + StandardScaler + PCA    | EVR 4e-6, loadings  cos  ≥ 0.99996, medians 5e-7, $\eta_{95} = 6$ identical                            |
| Correlation             | scipy.stats.pearsonr / spearmanr per pair, pairwise deletion                                                             | 5e-5 (4-dp rounding); row counts exact; symmetric                                                      |
| XGBoost                 | Reloaded the artifact on the regenerated held-out set; sklearn metrics                                                   | MAE 0.1056, P/R/F1 at 1e-4 and 1e-6, baselines and PCA(6/7) → XGB all reproduced to ≤ 5e-5             |
| TreeSHAP                | pred_contribs sum vs output_margin; own permutation importance                                                           | additivity 7e-6; ranking $\rho = 0.96$ , same top factor (older TLE age)                               |
| Foster Pc               | scipy.integrate.dblquad, 6 geometries (offset, rotated, AR 20, HBR > $\sigma$ , 1e-9 tail)                               | rel ≤ 1e-14                                                                                            |
| B-plane / covariance    | Own RTN covariances + QR-built encounter basis; 3-D Monte Carlo ( $4 \cdot 10^6$ ) of the combined error                 | Pc identical (rel 1e-15); MC within $0.5\sigma$                                                        |
| Chan                    | scipy.stats.ncx2 (isotropic, exact) and dblquad (anisotropic)                                                            | exact case 1e-16; approximation 0.2–2 % (26 % for HBR = $2\sigma_{\min}$ , as expected)                |
| Alfano / MC             | Numerical maximisation of dblquad over the covariance scale; own scipy MC (different seed)                               | Alfano 2e-9 rel; MC within $4\sigma$ ; MC refuses below its resolution                                 |
| Screening               | 60 real catalogue conjunctions: raw sgp4 brute force (1 s, then 1 ms scan)                                               | TCA 1.8 ms, miss 0.05 m                                                                                |
| Completeness            | 150-object shell, all pairs, sgp4 every 10 s with every local minimum refined                                            | 51 / 51 pairs found, 0 missed, 0 extra                                                                 |
| Frames                  | Vallado Alg. 15 GMST (Example 3-5); frontend computeGmst run in node                                                     | 8e-8 deg; sign convention correct                                                                      |
| CW / Hohmann / vis-viva | HCW ODE by DOP853; nonlinear two-body DOP853; vis-viva derivation                                                        | CW vs nonlinear 0.005 %; Hohmann 1e-15; transfer lands on $r_2$ (3e-10 km), e after $\Delta v_2$ 4e-14 |
| J2 propagators          | DOP853 with own J2 potential gradient                                                                                    | production RK4: 0.44 m after 1 day                                                                     |
| NASA SBM                | Hand EMR/M; KS tests for Lc, A/M mixture and $\Delta V$ ; np.interp re-implementation of the Johnson 2001 eq. 6–7 tables | all KS $p > 0.3$ ; tables 2e-4; momentum re-centred to 1e-4; no hyperbolic fragments                   |
| Fragment propagation    | DOP853 with own J2 + co-rotating drag; Vallado Table 8-4 retyped                                                         | 19 m after 6 h (≤ 1 % of the fragment $\sigma$ ); energy drift 1.2e-7 per day                          |
| Propulsion              | solve_ivp of dm/dt, dv/dt with a terminal event; datasheets; jet efficiency $F \cdot I_{sp} \cdot g_0 / 2P$              | 1e-13 rel; all 12 engines within datasheet values; electric efficiencies 0.48–0.65                     |
| Decision                | Own re-implementation of the HOW_IT_WORKS §11 formula on 500 random alerts; Pc sweep                                     | 0 mismatches; monotone; weights sum to 1; 1e-4 = CARA threshold                                        |
| Manoeuvres              | 18 planner options on 6 real alerts, each executed with apply_delta_v and then re-screened                               | miss 0.46 m, TCA 35 $\mu$ s, Pc 0.005 decades                                                          |

## Limitations stated by the verifier

- **TEME frame.** SGP4 outputs TEME, and the code treats TEME as inertial. Relative geometry and Pc are invariant to the rigid TEME→GCRF rotation (verified, FRM-6). The  $\sim 0.37^\circ$  of precession since J2000 only shifts *absolute* positions, by about 44 km. TEME→ECEF by GMST82 is the defined transform (Vallado 2006). UT1–UTC and polar motion are ignored, which costs at most 0.47 km on display. skyfield and astropy are not installed, so this bound is analytic.
- **Engine datasheet values** were compared against public figures recalled from manufacturer datasheets and papers, not fetched live. The per-engine notes give the ranges.
- **Covariance and HBR** are modelled (TLE age and object class), not taken from CDMs. The verification proves the maths is implemented correctly, not that the inputs are realistic.
- **SBM.** For fragments between 10 and 11 cm the bimodal (> 11 cm) A/M table is used, without the 8–11 cm bridging function. This affects the A/M distribution only.
- **Chan series.** This is an approximation for anisotropic covariances. It reaches 26 % relative error when HBR  $\approx 2\sigma_{\min}$ , and this is reported, not hidden. Foster remains the authoritative value.
- **Manoeuvre check.** It used executed burns on a cloned propagator, not the live :8000 server.

## Test suite (backend/tests)

| File                                 | What it proves                                                                                                   |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------|
| test_screening_real.py               | TCA/miss vs brute-force 1 s + 1 ms scan; far-now-close-later pair; Foster = dblquad; $\sigma$ grows with age     |
| test_debris_real.py                  | 40 J/g threshold; SBM count and size law; no hyperbolic fragments; momentum; energy drift; Vallado density       |
| test_cascade_real.py                 | BFS depth; $P(\text{hit}) = 1 - \prod(1 - P_c)$ ; re-propagated manoeuvre reaches $P_c < 1e-6$ ; SATCAT agencies |
| test_avoidance_real.py               | naive min- $\Delta v$ burn hits a third object → rejected; CW < 0.5 %; Tsiolkovsky vs RK4 1e-6                   |
| test_burn_real.py                    | zero burn moves nothing; predicted post-burn miss = executed (< 10 m)                                            |
| test_ml_real.py / test_ml_explain.py | beats baseline; MAE < 0.4 dex on fresh encounters; SHAP additivity; never touches physics Pc                     |
| test_decision_real.py                | monotone score; weights; thresholds; Chan $\approx$ Foster; MC within $4\sigma$ ; validation endpoint all-pass   |

# 11

## Assumptions, limitations & how to verify

what we model, what we assume, what we do not claim

- ▶ **No measured covariance.** TLEs have none;  $\sigma$  grows linearly with TLE age ( $\sigma_0$  Flohrer 2008, growth Vallado & Cefola 2012). With the bundled snapshot (epochs April-May 2026) TLEs are months old at the simulation date, the ellipse becomes huge and Pc collapses towards tiny values (dilution) — such alerts are flagged stale\_tle. Live Space-Track/CDN TLEs give meaningful Pc.
- ▶ **Masses are assumed.** SATCAT has no masses: scenario config → RCS class (50/500/2000 kg) → type default → 500 kg, tagged mass\_source. The fragment count scales with  $M^{0.75}$ .
- ▶ **Frames.** SGP4 output is TEME and is used as ECI (TEME  $\approx$  ECI, sub-km at these timescales); displayed altitude is  $|r| - 6371$  km (spherical Earth).
- ▶ **Breakup is one Monte-Carlo draw** (seed recorded per event), with an exponential atmosphere and no solar-activity-dependent density; at most 1000 fragments are propagated with weights.
- ▶ **Manoeuvres are impulsive.** Finite-burn validity is reported, not simulated; simultaneous burns of different alerts are not screened against each other; cascade check propagates fragments without drag.
- ▶ **ML is synthetic-trained.** Labels are physics Pc on generated geometry, not real CDMs; MAE degrades to 0.3372 dex when radial miss / altitude are missing. The trajectory RNN underperforms linear extrapolation and is marked experimental; GAT/GNN rankers are advisory only and operator feedback trains nothing.
- ▶ **Cascade  $\neq$  Kessler model.** The graph spans predicted encounters inside the screening window, not a multi-year population evolution. P(hit) assumes independent encounters.
- ▶ **Policy thresholds.** Pre-flight gates ( $> 200$  km trajectory clearance,  $> 60$  min TCA window) and the PREPARE/MONITOR tiers (1e-5, 1e-7) are our policy choices; only 1e-4 comes from NASA CARA / ESA practice.

### How to verify in five minutes

#### Commands

```
cd backend && set ORBIT_SENTINEL_SKIP_DOTENV=1 && python -m pytest -q
python -m app.ml.train_risk_surrogate # reproduces the model + card (seeded)
python docs/report/build_report.py # rebuilds this report from live data
```

| Endpoint                                   | What you will see                                                               |
|--------------------------------------------|---------------------------------------------------------------------------------|
| GET /api/physics/validation                | all live cross-checks with formula, values, error, tolerance                    |
| GET /api/analytics/model                   | correlation, PCA, PCA-vs-raw, SHAP, model card                                  |
| GET /api/model-metrics                     | held-out metrics of every model incl. the weak ones                             |
| GET /api/alerts                            | alerts with pc_checks, ml.contributions, decision, recommended_maneuver.options |
| POST /api/simulate {"name":"cascade_demo"} | computed-crossing scenario found by screening                                   |
| POST /api/debris/simulate                  | SBM breakup at predicted TCA + fragment alerts                                  |
| GET /api/physics/engines · /api/agencies   | engine catalogue with sources · SATCAT agency attribution                       |